For full credit explain your reasoning, showing all relevant work.

Exercise 1. Let G be an abelian group such that $|G| = 120$ and that G contains exactly 3 elements of order 2. Find a group H of the form

$$H = \mathbb{Z}_{p_1^{r_1}} \times \mathbb{Z}_{p_2^{r_2}} \times \cdots \times \mathbb{Z}_{p_k^{r_k}}$$

where $p_1, \ldots, p_k$ are (not necessarily different) prime numbers, such that $G \cong H$. Justify your answer.

Exercise 2. According to Theorem 18.1 every abelian group of order 270 is isomorphic to one of the following groups:

- $\mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_{27}$
- $\mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_9 \times \mathbb{Z}_3$
- $\mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3$

Which of these groups is isomorphic to $\mathbb{Z}_{270}$? Justify your answer.

Exercise 3. The goal of this exercise is to prove the converse of Theorem 18.2 from the lecture notes. Assume that G, G_1, G_2 are abelian groups such that $G \cong G_1 \times G_2$. Show that there exist subgroups $H_1, H_2 \subseteq G$ satisfying the following conditions:

1) $H_1 \cap H_2 = \{e\}$.
2) For any $g \in G$ there exist $h_1 \in H_1$ and $h_2 \in H_2$ such that $g = h_1 h_2$.
3) $H_1 \cong G_1$ and $H_2 \cong G_2$.

PRACTICE PROBLEMS

Exercises below are for practice only - do not turn them in for grading.

Practice Exercise 1. Show that, up to isomorphism, there are two abelian groups of order 108 that have exactly 2 elements of order 3.

Practice Exercise 2. Let $p_1, p_2, \ldots, p_{100}$ be different primes. Up to isomorphism, how many different abelian groups are there of order $p_1^4 \cdot p_2^4 \cdot \ldots \cdot p_{100}^4$? Justify your answer.

Practice Exercise 3. a) Show that every abelian group of order 45 contains an element of order 15.

b) Give an example of an abelian group of order 45 which has no element of order 9. Justify your answer.

Practice Exercise 4. Let G be an abelian group of order n which has exactly one subgroup of order d for each number d which divides n. Show that G is a cyclic group.